

TABLE II
DC Electrical Characteristics (Continued)

(Note 3)			LF155/6			LF256/7 LF356B			LF355/6/7			Units
Symbol	Parameter	Conditions	Min.	Typ.	Max.	Min.	Typ.	Max.	Min.	Typ.	Max.	
I_B	Input bias current	$T_J = 25^\circ\text{C}$ (Notes 3, 5) $T_J \leq \text{High}$	—	30	100	—	30	100	—	30	200	pA nA
R_{IN}	Input resistance	$T_J = 25^\circ\text{C}$	—	10^{12}	—	—	10^{12}	—	—	10^{12}	—	Ω
A_{VOL}	Large signal voltage gain	$V_S = \pm 15\text{V}$, $T_J = 25^\circ\text{C}$ $V_S = \pm 10\text{V}$, $R_L = 2\text{k}$ Over temperature	50	200	—	50	200	—	25	200	—	V/mV V/mV
V_O	Output voltage swing	$V_S = \pm 15\text{V}$, $R_L = 10\text{k}$ $V_S = \pm 15\text{V}$, $R_L = 2\text{k}$	± 12 ± 10	± 13	—	± 12 ± 10	± 13	—	± 12 ± 10	± 13	—	V V
V_{CM}	Input common-mode voltage range	$V_S = \pm 15\text{V}$	± 11	+15.1 -12	—	± 11	+15.1 -12	—	+10	+15.1 -12	—	V V
CMRR	Common-mode rejection ratio	—	85	100	—	85	100	—	80	100	—	dB
PSRR	Supply voltage rejection ratio	—	85	100	—	85	100	—	80	100	—	dB

dependent. Differential input voltage is another related parameter. For some op-amps, manufacturers recommend use of resistors in series with the two inputs in order to limit the input current if the differential input voltage rating were exceeded. High-speed op-amp AD 829 is an example.

Power dissipation rating is typically specified at an ambient temperature of 25°C and it must be de-rated for operation at elevated temperatures as per the specified de-rating curve or for given values of maximum allowable junction temperature T_J (max) and junction-to-ambient thermal resistance θ_{JA} in the absence of any de-rating curve.

Electrical characteristics

Electrical characteristics are categorised by some manufacturers as DC and AC. Others specify all under a common heading. Key DC characteristics include input bias current, input offset

Dependence of key parameters on operating temperature, load resistance and supply voltages should not be ignored while designing an application circuit.

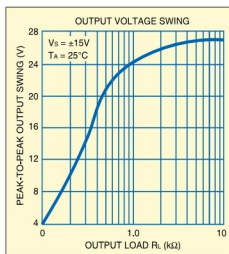


Fig. 6: Output voltage swing versus load resistance for LF156

current, input offset voltage, offset drifts with temperature, open-loop gain, input resistance, output voltage swing, input common-mode voltage range, CMRR and PSRR. Gain-bandwidth product, slew rate and settling time are the major AC characteristics.

Each of the electrical characteristics is specified under certain test conditions. These test conditions should not be ignored while designing an application circuit. The specified values will be different under different conditions and in some cases the variation could be significant. As an illustration, while operating op-amp LF156 on $\pm 15\text{V}$

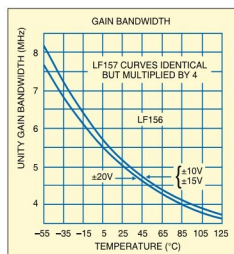


Fig. 7: Output voltage swing as a function of operating temperature for LF156

supply, peak-to-peak output voltage swing is specified to be 24V for a load resistance of 2 kilo-ohms (Table II). The parameter falls rapidly for lower values of load resistance—as is evident from the output voltage swing versus load resistance curve given in the datasheet (Fig. 6).

Output voltage swing is also a strong function of operating temperature (see Fig. 7).

As another illustration, response time parameter is observed to be strongly dependent upon input overdrive. For LM 119/219/319, it improves from about 100 ns to 50 ns as